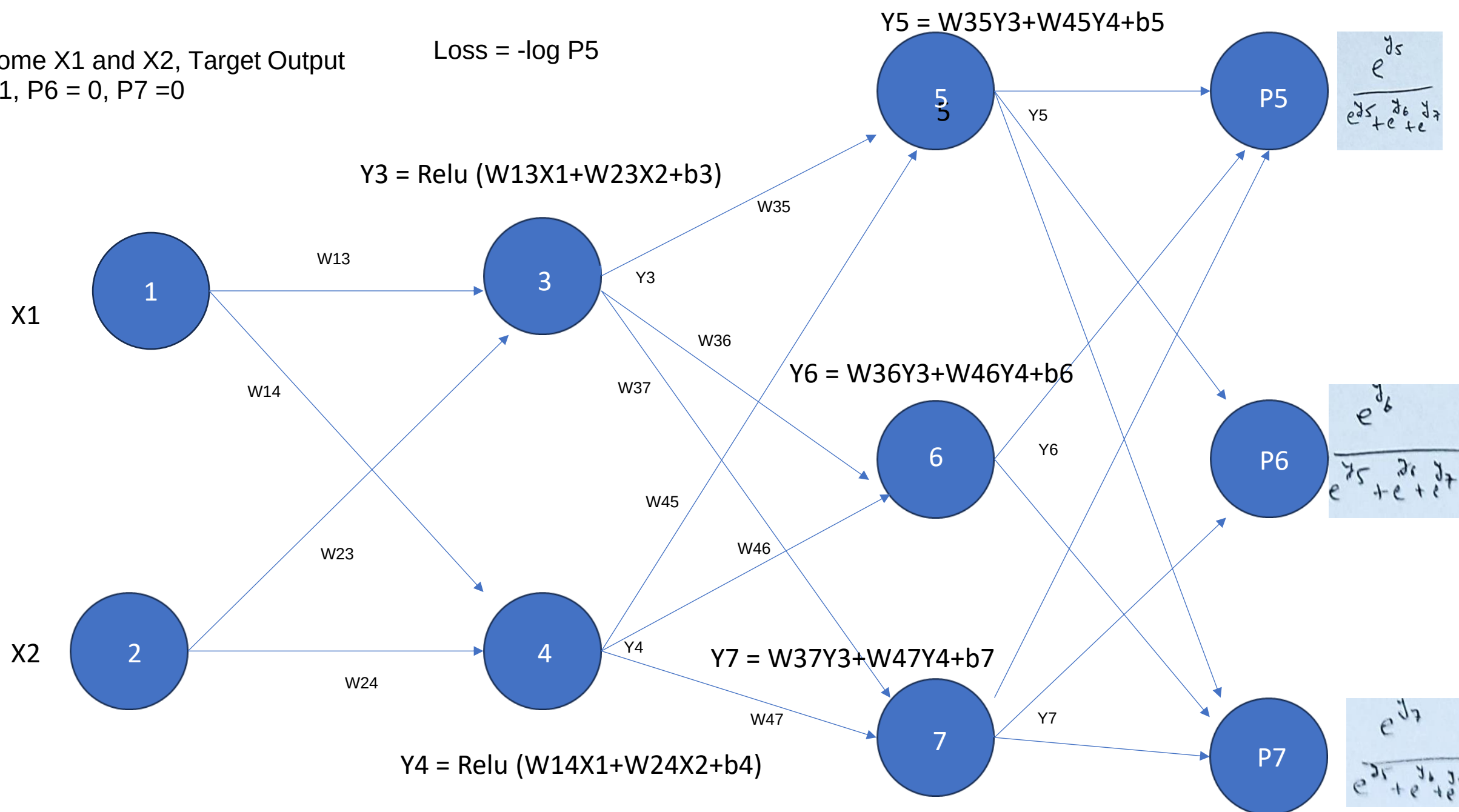


For some X1 and X2, Target Output
P5 = 1, P6 = 0, P7 = 0

Loss = -log P5



Parameter Update 2

Chowdhury Mofizur Rahman

April 23, 2025

Assume that the target class is at node 5. At first start with the output nodes. Define loss function and differentiate loss function with respect to all the output nodes.

$$\begin{aligned} Loss &= -\log P_5 \\ \frac{dloss}{dP_5} &= -\frac{1}{P_5} \\ \frac{dLoss}{dY_5} &= \frac{dLoss}{dP_5} \cdot \frac{dP_5}{dY_5} = -\frac{1}{P_5} \cdot P_5(1 - P_5) = P_5 - 1 \\ \frac{dLoss}{dY_6} &= \frac{dLoss}{dP_5} \cdot \frac{dP_5}{dY_6} = -\frac{1}{P_5} \cdot (-P_5 \cdot P_6) = P_6 \\ \frac{dLoss}{dY_7} &= \frac{dLoss}{dP_5} \cdot \frac{dP_5}{dY_7} = -\frac{1}{P_5} \cdot (-P_5 \cdot P_7) = P_7 \end{aligned}$$

Write the expressions for output nodes

$$Y_5 = W_{35}Y_3 + W_{45}Y_4 + b_5$$

$$Y_6 = W_{36}Y_3 + W_{46}Y_4 + b_6$$

$$Y_7 = W_{37}Y_3 + W_{47}Y_4 + b_7$$

Now you can differentiate loss function with respect to all the output layer parameters $W_{35}, W_{45}, W_{36}, W_{46}, W_{37}, W_{47}, b_5, b_6, b_7$ using chain rule

$$\begin{aligned} \frac{dLoss}{dW_{35}} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{dW_{35}} = (P_5 - 1) \cdot Y_3 \\ \frac{dLoss}{dW_{45}} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{dW_{45}} = (P_5 - 1) \cdot Y_4 \\ \frac{dLoss}{dW_{36}} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{dW_{36}} = P_6 \cdot Y_3 \\ \frac{dLoss}{dW_{46}} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{dW_{46}} = P_6 \cdot Y_4 \\ \frac{dLoss}{dW_{37}} &= \frac{dloss}{dY_7} \cdot \frac{dY_7}{dW_{37}} = P_7 \cdot Y_3 \\ \frac{dLoss}{dW_{47}} &= \frac{dloss}{dY_7} \cdot \frac{dY_7}{dW_{47}} = P_7 \cdot Y_4 \\ \frac{dLoss}{db_5} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{db_5} = P_5 - 1 \\ \frac{dLoss}{db_6} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{db_6} = P_6 \\ \frac{dLoss}{db_7} &= \frac{dloss}{dY_7} \cdot \frac{dY_7}{db_7} = P_7 \end{aligned}$$

Now using gradient descent approach the update equations for all the output layer parameters are:

$$W_{35} = W_{35} - \eta.Y_3.(P_5 - 1)$$

$$W_{45} = W_{45} - \eta.Y_4.(P_5 - 1)$$

$$W_{36} = W_{36} - \eta.Y_3.P_6$$

$$W_{46} = W_{46} - \eta.Y_4.P_6$$

$$W_{37} = W_{37} - \eta.Y_3.P_7$$

$$W_{47} = W_{47} - \eta.Y_4.P_7$$

$$b_5 = b_5 - \eta.(P_5 - 1)$$

$$b_6 = b_6 - \eta.P_6$$

$$b_7 = b_7 - \eta.P_7$$

Now consider the hidden nodes 3 and 4. Write the expressions for Y_3 and Y_4 .

$$Y_3 = Relu\{W_{13}.X_1 + W_{23}.X_2 + b_3\}$$

$$Y_4 = Relu\{W_{14}.X_1 + W_{24}.X_2 + b_4\}$$

Now differentiate the Loss function with respect to hidden outputs Y_3 and Y_4 . Both nodes 3 and 4 have parents 5, 6 and 7. Following the chain rule

$$\frac{dLoss}{dY_3} = \frac{dloss}{dY_5} \cdot \frac{dY_5}{dY_3} + \frac{dloss}{dY_6} \cdot \frac{dY_6}{dY_3} + \frac{dloss}{dY_7} \cdot \frac{dY_7}{dY_3} = (P_5 - 1).W_{35} + P_6.W_{36} + P_7.W_{37}$$

$$\frac{dLoss}{dY_4} = \frac{dloss}{dY_5} \cdot \frac{dY_5}{dY_4} + \frac{dloss}{dY_6} \cdot \frac{dY_6}{dY_4} + \frac{dloss}{dY_7} \cdot \frac{dY_7}{dY_4} = (P_5 - 1).W_{45} + P_6.W_{46} + P_7.W_{47}$$

Now we can differentiate Loss function with respect to all the hidden layer parameters W_{13} , W_{23} , W_{14} , W_{24} , b_3 and b_4

$$\frac{dLoss}{dW_{13}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{13}} = \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3).X_1$$

$$\frac{dLoss}{dW_{23}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{23}} = \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3).X_2$$

$$\frac{dLoss}{dW_{14}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{14}} = \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4).X_1$$

$$\frac{dLoss}{dW_{24}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{24}} = \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4).X_2$$

$$\frac{dLoss}{db_3} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{db_3} = \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3)$$

$$\frac{dLoss}{db_4} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{db_4} = \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4)$$

Therefore the updated equations for all the hidden layer parameters are as follows:

$$W_{13} = W_{13} + \eta \cdot \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3).X_1$$

$$W_{23} = W_{23} + \eta \cdot \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3).X_2$$

$$W_{14} = W_{14} + \eta \cdot \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4).X_1$$

$$W_{24} = W_{24} + \eta \cdot \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4).X_2$$

$$b_3 = b_3 + \eta \cdot \{(P_5-1).W_{35}+P_6.W_{36}+P_7.W_{37}\}.Relu'(W_{13}.X_1+W_{23}.X_2+b_3)$$

$$b_4 = b_4 + \eta \cdot \{(P_5-1).W_{45}+P_6.W_{46}+P_7.W_{47}\}.Relu'(W_{14}.X_1+W_{24}.X_2+b_4)$$

$$\begin{aligned}
 \frac{dP_5}{dy_5} &= \frac{d}{dy_5} \cdot \frac{e^{y_5}}{e^{y_5} + e^{y_6} + e^{y_7}} \\
 &= \frac{d}{dy_5} \cdot e^{y_5} (e^{y_5} + e^{y_6} + e^{y_7})^{-1} \\
 &= e^{y_5} \cdot \frac{d}{dy_5} (e^{y_5} + e^{y_6} + e^{y_7})^{-1} \\
 &\quad + (e^{y_5} + e^{y_6} + e^{y_7})^{-1} \cdot \frac{d}{dy_5} e^{y_5}
 \end{aligned}$$

$$= - \frac{e^{y_5} \cdot e^{y_5}}{(e^{y_5} + e^{y_6} + e^{y_7})^2} + \frac{e^{y_5}}{(e^{y_5} + e^{y_6} + e^{y_7})}$$

$$= - \frac{e^{y_5}}{e^{y_5} + e^{y_6} + e^{y_7}} \left(\frac{e^{y_5}}{e^{y_5} + e^{y_6} + e^{y_7}} - 1 \right)$$

$$= -P_5 (P_5 - 1) = P_5 (1 - P_5)$$

$$\begin{aligned}
 \frac{dP_5}{dy_6} &= \frac{d}{dy_6} e^{y_5} (e^{y_5} + e^{y_6} + e^{y_7})^{-1} \\
 &= - \frac{e^{y_5} \cdot e^{y_6}}{(e^{y_5} + e^{y_6} + e^{y_7})^2}
 \end{aligned}$$

$$= - \frac{e^{y_5}}{e^{y_5} + e^{y_6} + e^{y_7}} \cdot \frac{e^{y_6}}{e^{y_5} + e^{y_6} + e^{y_7}} = -P_5 \cdot P_6$$

$$\text{Similarly } \frac{dP_5}{dy_7} = -P_5 \cdot P_7$$